



CVE Explorer

Real-time Vulnerability Intelligence • AI-Powered Remediation • Multi-Source Data

Or Davidovitz • Alexandra Belkind • Peleg Lampl

Upwind Bootcamp 2026

The Problem



Thousands of new CVEs daily

Security teams can't keep up with the volume of vulnerabilities being published every day



Fragmented data sources

NVD, GitHub, CVE.org — each has different data. No single tool aggregates them intelligently



No remediation guidance

Knowing about a vulnerability is useless without knowing how to fix it



Slow response time

By the time teams understand a CVE, attackers have already exploited it

We built CVE Explorer to solve this →

System Architecture



Data Layer

Alexandra Belkind

Scrapes CVEs from NVD, GitHub
Advisory DB & CVE.org
Merges into enriched SQLite database
619+ unique CVEs



API Layer

Or Davidovitz

FastAPI REST backend
10+ endpoints with search, filter, sort
AI-powered remediation via Azure
OpenAI



UI Layer

Peleg Lampl

React + Vite frontend
Live search, severity filters, sorting
CVE detail modal with AI fix button

Each layer is independent — agreed on API contract upfront, built in parallel

Data Layer

Alexandra Belkind

619+

CVEs collected

NVD

National Vulnerability Database

CVSS scores, vendor/product info, CPE data

GHSA

GitHub Advisory Database

Affected packages across npm, PyPI, Maven, Go

CVE.org

MITRE CVE Services

Authoritative descriptions and references

- ✓ Upsert logic — re-running is safe, no duplicates
- ✓ Incremental scraping — only fetch changes since last run
- ✓ Multi-source merge — richest combined view per CVE

Backend / API Layer

FastAPI · Python

Or Davidovitz

Mock data fallback



API works without a DB — all 3 devs built in parallel from day one

AI remediation



Azure OpenAI GPT-5.4 generates actionable fix recommendations per CVE

AI Caching



Same CVE never hits AI twice — saves cost, instant on repeat views

Multi-source filter



Filter CVEs by source: NVD, GHSA, CVE.org

DB change detection



MD5 hash detects new data — POST /refresh, no restart needed

Smart search



Covers ID, title, description, vendors, products AND packages

API Endpoints

GET

`/cves`

Search, filter by severity/source, sort, paginate

GET

`/cves/{id}?remediation=true`

Single CVE + AI-generated fix recommendation

GET

`/cves/{id}/similar`

Related vulnerabilities by keyword matching

GET

`/cves/{id}/explore`

Live cross-reference: NVD, CVE.org, MITRE, GitHub

GET

`/cves/trending`

Most recent dangerous CVEs

GET

`/cves/timeline`

CVEs grouped by month — ready for charts

GET

`/stats`

Breakdown by severity AND by source

POST

`/refresh`

Detect DB changes, clear AI cache, serve fresh data

GET

`/health`

Uptime, DB hash, AI cache status

✦ AI-Powered Remediation

The standout feature — no other CVE tool in the class has this

1



User opens CVE

Clicks any vulnerability
in the list

2



**Clicks Get AI
Remediation**

One button in the
detail modal

3



**Backend calls Azure
OpenAI**

GPT-5.4 receives CVE
description + severity

4



AI generates fix

3-5 actionable bullet
points, security-expert
level

5



Result is cached

Same CVE = instant
response on repeat
views

"When a user asks how to fix CVE-2024-XXXX, they get expert-level guidance in seconds — not hours of research"

Frontend / UI Layer

Peleg Lampl

React • Vite

Live Dashboard



Real-time counts by severity — CRITICAL, HIGH, MEDIUM, LOW

Filter & Sort



Filter by severity, sort by date/score/alphabetically

AI Remediation Button



One click to get AI-powered fix recommendations

Loading Skeletons



Smooth UX with animated placeholders while data loads

Smart Search



Search by CVE ID, keyword, vendor or technology with debounce

Detail Modal



Full CVE info, CVSS score bar, published date, NVD link

Ask GPT Button



Opens ChatGPT with full CVE context pre-loaded

Beautiful Design



Custom beach theme, smooth animations, responsive layout

Challenges We Faced

Or — Backend

⚡ FastAPI route ordering

/cves/trending was being matched by /cves/{id} before reaching its own handler. Fixed by placing specific routes before dynamic ones — a key lesson in API design.

💡 *Specific routes must always come before dynamic ones*

Alexandra — Data

⚡ NVD rate limiting

The NVD public API allows only ~5 requests per 30 seconds without an API key. Solved with 6-second pauses between pages and automatic retry logic on 429 errors.

💡 *Always design scrapers with rate limits in mind*

Peleg — Frontend

⚡ API response format mismatch

The frontend expected data directly but the API returned {total, data: [...]}. Fixed with flexible parsing: `Array.isArray(data) ? data : data.data || data.cves`

💡 *Agree on API contracts before building — not after*

What We Built

- ✓ 619 real CVEs from 3 sources: NVD, GitHub Advisory, CVE.org
- ✓ FastAPI backend with 10+ endpoints — search, filter, sort, paginate
- ✓ AI-powered remediation with caching via Azure OpenAI GPT-5.4
- ✓ Beautiful React UI — live search, severity filters, CVE detail modal
- ✓ DB change detection with MD5 — always fresh data, no restart needed
- ✓ Docker support — entire system with one docker-compose up
- ✓ End-to-end working system — data → API → UI

Python · FastAPI · SQLite · React · Vite · Azure OpenAI · Docker